David Nguyen

604.307.3521 | thentd2003@gmail.com |

EXPERIENCE

Analytic Systems Sept. 2023 – Dec. 2024
Engineering Co-op Delta, BC

- Designed and built a **high-voltage flyback converter** (24VDC input) as an independent design project – first end-to-end power converter design – and built custom test jigs to bench-validate output voltage and efficiency targets.
- Performed **EMC/EMI compliance testing** (conducted 10kHz–10MHz, radiated 30–300MHz) on a batch of **50 power inverters** using a shielded RF chamber, LISN, and biconical antenna.
- Diagnosed and resolved the sole non-conforming unit out of 50, ensuring full-batch compliance and on-time shipment with documented conformance results.
- Supported broader converter/inverter troubleshooting and power electronics **R&D** for industrial, marine, and defense-grade product lines.

Tech Maple Works Dec. 2025 – Jul. 2026
Engineering Consultant – Power Electronics Burnaby, BC

- Architected a **half-bridge LCLR series-resonant induction heater** based on the "direct-drive" topology, defining the full power signal chain from DC bus through half-bridge switching, matching inductor, tank capacitor bank, and series-resonant work coil.
- Diagnosed a **gate-driver bandwidth limitation** causing destructive hard-switching; retargeted operating frequency to 57–70kHz and resolved distortion with a low-cost Schottky-diode fast-turnoff network in place of a full totem-pole buffer.
- Iteratively re-sized the tank's matching inductor and redesigned the capacitor bank into a split parallel arrangement, correcting a load-flip failure mode and reducing thermal stress on individual components.
- Diagnosed **Curie-point-driven resonant drift** as the root cause of the open-loop design's fragility, and architected a **closed-loop PLL control system** to track resonance in real time.

SFU Rocketry Team Sept. 2022 – Apr. 2023
Power Team Member Surrey, BC

- Designed and developed stable **5V and 3.3V DC-DC converters** for the rocket's onboard power distribution system, powering avionics and communication subsystems.
- Built a **dynamic load testing system** and simulated converter designs in **Altium and LTspice** prior to hardware validation.
- Integrated completed power modules into the rocket's overall power system, coordinating with subsystem teams to meet reliability requirements ahead of launch.

EDUCATION

Simon Fraser University 2021 – Present
Bachelor of Applied Science in Engineering Science (Electronics) Burnaby, BC

Simon Fraser University May 2023 – Sep 2023
Siemens Mechatronic Systems Certification Program (Level 1) Burnaby, BC

SKILLS & INTERESTS

Hardware & Lab : Soldering, Oscilloscope & DMM, PCB Design & Fabrication, Circuit Troubleshooting, 3D Printing
Design & Simulation : Altium Designer, LTspice, KiCad, OrCAD, Proteus, MATLAB/Simulink, SolidWorks
Programming & HDL : C++, Python, VHDL, Verilog, SystemVerilog, Xilinx Vivado, Intel Quartus, ModelSim, GNU Radio
Languages : English (Proficient), Vietnamese (Native)